Many Daughters Pre-analysis Plan

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1 Introduction

We study how having daughters influences parents' political views and their behaviour and attitudes surrounding gender equality, using data from the German Socio-Economic Panel (SOEP). This will be one of multiple studies addressing the same hypotheses, which will later be compared and combined into a larger meta-science project.

Recent literature suggests that having daughters may play an important role in the behaviour and attitudes of parents. In a UK sample, rearing daughters was found to decrease the likelihood that fathers would hold traditional attitudes about gender roles (Borrell-Porta et al., 2019) and in a Danish sample, having a daughter increased the chance that managers would substitute males for comparable females when hiring (Ronchi and Smith, 2024). We extend this work by examining in a German general population sample whether having a daughter leads to more household gender equality, support for gender equality more broadly, particularly in fathers, and whether having a daughter has spillovers to parents' political views.

To address these questions, we employ a natural experiment in which we study parents' attitudes and behaviours when they have daughters compared to those who do not have daughters. To control for any self-selection bias of those who chose to have children versus those who didn't, we take as our control group the parents who only had male children, as compared to our treatment group of parents with at least one daughter.

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2 Hypotheses

We test four hypotheses:

H1: Having daughters leads to stronger support for gender equality.

H2: Having daughters leads to more left-leaning political preferences.

H3: Having daughters leads to more gender equal behaviour/outcomes in the household.

H4: Having daughters leads to a larger increase in the support of gender equality for men than for women.

3 Variables

We will utilise a 5% sample of the data from the German Socio-Economic Panel (SOEP); a longitudinal representative survey panel of private households in Germany with up to 90,357 individual respondents between 2000 and 2020. The panel is made up of a series of random probability samples based on a nationwide two-stage stratified sampling procedure. Responses are collected via mostly face-to-face interviews with all household members of ages 16 and over.[1] Though the core SOEP panel comprises both household data provided by the head of each household and individual-level data from each household member, we focus on individual level data since we expect household data to be skewed towards those who self-selected into being the perceived household 'head' and hence would not be representative of all parents.

3.1 Independent Variables

As our primary independent variable for all hypotheses, we use a single dummy variable for whether individual respondents have one or more daughters (1) or whether they have only sons (0). We construct the dummy from the variables kidsex01 - kidsex19 in the biobirth dataset by coding it as '1' if the respondent answers to one or more of the kidsex variables that the sex of their child is 'Female', and '0' if the respondent answers to one or more of the kidsex variables that the sex of their child is 'Male' and never answers that the sex of their child is 'Female'. As discussed above, this limits our comparison to only those who chose to have children and excludes childless respondents.

As secondary independent variables for our interaction terms and to test H4, we also consider a pre (0) and post (1) treatment dummy coded as 0 before respondents have a daughter and 1 after respondents have a daughter, as well as a gender dummy depending on whether the respondent is Male (0) or Female (1). This will come from the pla0009_h variable in the pl dataset.

[1] See `https://www.diw.de/en/diw_02.c.299052.en/survey_methods.html#299746` for more details on the survey methods and sample.

3.2 Dependent Variables

As our first dependent variable to test both H1 and H4, we consider answers to the SOEP core question: "It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally" as our measure of support for gender equality (variable plh0308_v2 in the pl dataset from the individual questionnaires, N = 25,570). Answers to this were provided on a 7-point Likert item from "Completely Disagree" (1) to "Fully agree" (7).

As our second dependent variable to test H2, we consider answers to the SOEP core question: "In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?" as our measure of political views (variable plh0004 in the pl dataset from the individual questionnaires, N = 90,972). Answers to this were provided on an 11-point Likert item from "On the far left" (0) to "Far right" (10).

As our third dependent variable to test H3, we consider answers to the SOEP core question: "How many hours per week are stipulated in your contract (excluding overtime)?" as our measure of gender equality in the household, by capturing whether both parents reduce working hours equivalently after having a child (variable plb0176_h in the pl dataset from the individual questionnaires, N = 38,9905). Answers to this were a continuous measure of work hours.

4 Inclusion/exclusion criteria

4.1 Inclusion criteria

- We include only families with 1-2 children. This will focus the sample and avoid generational gaps for families.
- We include data limited to the years 2000 to 2020.
- In some survey years, there are rare cases of younger participants who have also completed this questionnaire. Hence, we include only responses from individuals over the age of 18.
- We aim to only include Computer-Assisted Personal Interviews (CAPI) compared to Paper-and-Pencil Interviews (PAPI) to avoid any experimenter effects.
- We include individuals living in private households within Germany.
- We also only include respondents who state they have children.

4.2 Exclusion criteria

- Any negative or unacceptable values for a variable will be considered as missing values. There are two possible origins of missing values: the

respondent's answer or the survey design. In the first case, the respondent may refuse to answer or not know the answer or may report invalid values. In the second case, the interview design may exclude respondents with certain characteristics from some questions (e.g., men will never be asked if they are pregnant). The codes for these types of responses are presented in the table below:

Table 1: Negative values within the dataset explaining the responses from the participants.

Code	Label
-1	No answer / don't know
-2	Does not apply
-3	Implausible value
-4	Inadmissable multiple response
-5	Not included in this version of the questionnaire
-6	Version of questionnaire with modified filtering
-7	Only available in less restricted edition
-8	Question not part of the survey program this year

Notes: The negative values are codes determining how the participant answered a question. The responses from these would be considered as missing values in the analysis.

- A variable will be excluded if the participant did not remain in Germany. If someone moves abroad, they exit the SOEP and so provide incomplete data across the panel. In these cases we exclude these observations.
- We do not consider any measurements that are not based on individual interviews, meaning that we do not include any household-level data for our main analyses.
- We also exclude variables that have built-in filters (e.g. pregnancy questions only for women) and special missing codes, with the single exception being participants who did not answer the question regarding contracted working hours. If a participant did not complete this question because they are, for example, a stay-at-home parent, they will receive a score of 0 on this variable.
- We exclude observations where respondents did not respond to every variable we consider in our analysis and waves where our variable of interest was not elicited.

5 Analysis

To test our first hypothesis that having daughters leads to stronger support for gender equality, we will conduct a staggered difference-in-difference estimation

with time and individual-level fixed effects and clustered standard errors of agreement that men and women should share household responsibilities equally (Y_it) on the dummy for having a daughter or not ($Daughter_{it}$), a dummy for pre and post having the first daughter ($Post_{it}$) and an interaction term of the two dummies ($Daughter_{it}$ x $Post_{it}$). We will also include an array of control variables for factors such as age, gender, income, nationality, the number of respondents' children, relationship status, child support received etc.

$$\begin{aligned} Y_{it} = \alpha &+ \beta_2 Daughter_{it} + \beta_3 Post_{it} \\ &+ \beta_4 (Daughter_{it} \times Post_{it}) \\ &+ \gamma Controls_{it} + \epsilon_{it} \end{aligned} \tag{1}$$

To test our second hypothesis that having daughters leads to more left-leaning political preferences, we will also conduct a staggered difference-in-difference estimation with time and individual-level fixed effects and clustered standard errors of self-reported political leaning (Y_it) on the dummy for having a daughter or not ($Daughter_{it}$), a dummy for pre and post having the first daughter ($Post_{it}$) and an interaction term of the two dummies ($Daughter_{it}$ x $Post_{it}$). We will also include the same array of control variables as the previous equation.

$$\begin{aligned} Y_{it} = \alpha &+ \beta_2 Daughter_{it} + \beta_3 Post_{it} \\ &+ \beta_4 (Daughter_{it} \times Post_{it}) \\ &+ \gamma Controls_{it} + \epsilon_{it} \end{aligned} \tag{2}$$

To test our third hypothesis that having daughters leads to more gender equality in the household, we will also conduct a staggered difference-in-difference estimation with time and household-level fixed effects and clustered standard errors. As our dependent variable, we will match up all male-female partners in our sample using household codes, then calculate the absolute difference in contracted working hours between the Male (father) respondent and the Female (mother) respondent (Y_it). Notice that by taking the absolute difference, we do not distinguish between differences in favour of either parent, but instead consider any type of unequal distribution of work. As our independent variables, we take the dummy for having a daughter or not ($Daughter_{it}$), a dummy for pre and post having the first daughter ($Post_{it}$) and an interaction term of the two dummies ($Daughter_{it}$ x $Post_{it}$). We will also include an array of control variables.

$$\begin{aligned} Y_{it} = \alpha &+ \beta_2 Daughter_{it} + \beta_3 Post_{it} \\ &+ \beta_4 (Daughter_{it} \times Post_{it}) \\ &+ \gamma Controls_{it} + \epsilon_{it} \end{aligned} \tag{3}$$

Finally, to test our fourth hypothesis that having daughters leads to a larger increase in the support of gender equality for men than for women, we will also

conduct a staggered difference-in-difference estimation with time and individual-level fixed effects and clustered standard errors. Our dependent variable will be agreement that men and women should share household responsibilities equally (Y_it), as in our test for hypothesis 1, but in addition to including the dummy for having a daughter or not ($Daughter_{it}$) and a dummy for pre and post having the first daughter ($Post_{it}$), we will also include a dummy for whether the respondent is Male or Female ($Male_{it}$) and a triple interaction term between the three dummies ($Daughter_{it}$ x $Post_{it}$ x $Male_{it}$). We will also include an array of control variables.

$$\begin{aligned} Y_{it} = \alpha &+ \beta_2 Daughter_{it} + \beta_3 Post_{it} + \beta_4 Male_{it} \\ &+ \beta_5 (Daughter_{it} \times Post_{it} \times Male_{it}) \\ &+ \gamma Controls_{it} + \epsilon_{it} \end{aligned} \tag{4}$$

6 Interpretation of results

Across all tests, we take α of 0.05 as our significance level.

To determine whether we find support for H1, we consider whether the interaction term in regression 1 is significant, with our measure of gender equality attitudes being higher (more supported) after having a first daughter compared to after having sons.

To determine whether we find support for H2, we consider whether the interaction term in regression 2 is significant, with our measure of political preferences being lower (more left-leaning) after having a first daughter compared to after having sons.

To determine whether we find support for H3, we consider whether the interaction term in regression 3 is significant, with the difference in contracted work hours between parents being smaller (more equal) after having a first daughter compared to after having a son.

Finally, to determine whether we find support for H4, we consider whether the triple interaction term in regression 4 is significant, with a larger effect of having a daughter rather than sons on the gender equality attitudes of Male respondents compared to Female respondents.

References

Borrell-Porta, M., J. Costa-Font, and J. Philipp (2019): "The 'mighty girl'effect: does parenting daughters alter attitudes towards gender norms?" *Oxford Economic Papers*, 71, 25–46.

Ronchi, M. and N. Smith (2024): "Daddy's girl: Daughters, managerial decisions, and gender inequality," *Working Paper*.